

Solutions to Exercise Series 2: Diodes and Zener Diodes

Fundamental Electronics

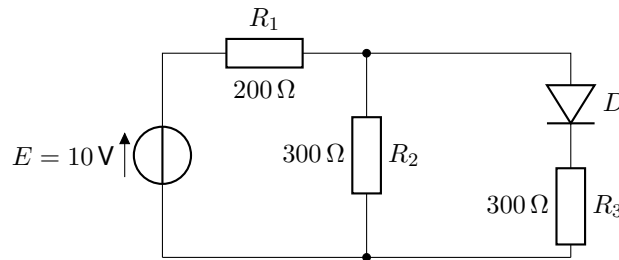
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Exercise 1: Diode State and Current Analysis

Assume the diode is ideal with a threshold voltage of $V_D = 0.7\text{ V}$.

Circuit 1(a)



Step 1: Assume the diode is OFF (Blocked).

If the diode is open, no current flows through R_3 . The voltage across the parallel branch (let's call it V_p) is simply a voltage divider between E , R_1 , and R_2 :

$$V_p = E \times \frac{R_2}{R_1 + R_2} = 10 \times \frac{300}{200 + 300} = 10 \times \frac{300}{500} = 6\text{ V}$$

Since the potential $V_p = 6\text{ V}$ is greater than the diode threshold voltage (0.7 V), our assumption is false. **The diode is ON (Conducting).**

Step 2: Calculate the current with the diode ON.

Replace the diode with a 0.7 V voltage drop. Apply Nodal Analysis at the top junction (voltage V_p):

$$\begin{aligned}\frac{E - V_p}{R_1} &= \frac{V_p}{R_2} + \frac{V_p - V_D}{R_3} \\ \frac{10 - V_p}{200} &= \frac{V_p}{300} + \frac{V_p - 0.7}{300}\end{aligned}$$

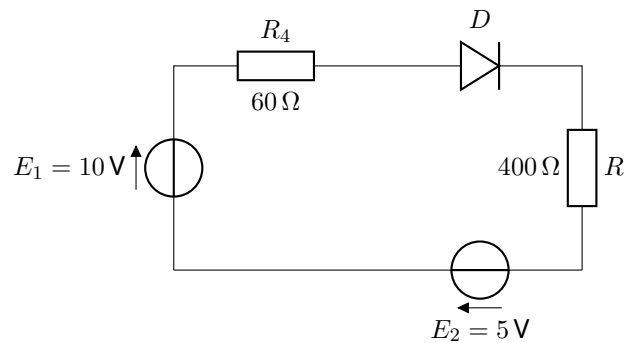
Multiply the entire equation by 600 to clear denominators:

$$\begin{aligned}3(10 - V_p) &= 2V_p + 2(V_p - 0.7) \\ 30 - 3V_p &= 2V_p + 2V_p - 1.4 \\ 31.4 &= 7V_p \implies V_p \approx 4.486\text{ V}\end{aligned}$$

The current flowing through the diode (I_D) is the current through the rightmost branch:

$$I_D = \frac{V_p - V_D}{R_3} = \frac{4.486 - 0.7}{300} = \frac{3.786}{300} \approx 0.01262\text{ A} = \mathbf{12.62\text{ mA}}$$

Circuit 1(b)



Step 1: Check Diode State.

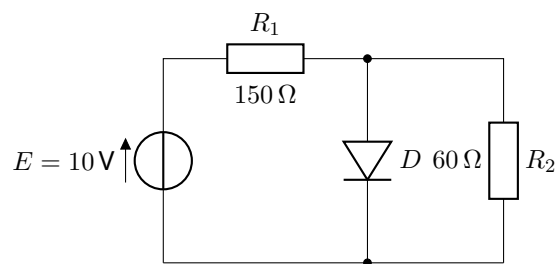
The two voltage sources oppose each other. E_1 (10 V) pushes current clockwise, while E_2 (5 V) pushes counter-clockwise. Since $10\text{ V} > 5\text{ V}$, the net voltage forces current clockwise, forward-biasing the diode. **The diode is ON.**

Step 2: Calculate the current.

Apply Kirchhoff's Voltage Law around the single loop:

$$\begin{aligned}E_1 - R_4 I - V_D - R I - E_2 &= 0 \\10 - 60I - 0.7 - 400I - 5 &= 0 \\4.3 &= 460I \\I &= \frac{4.3}{460} \approx 0.00935\text{ A} = \mathbf{9.35\text{ mA}}\end{aligned}$$

Exercise 2: Clamping/Parallel Diode Circuit



1. Determine Diode State:

Assume the diode is OFF. The voltage across its terminals would be determined by the voltage divider of R_1 and R_2 :

$$V_{open} = 10 \times \frac{60}{150 + 60} = 10 \times \frac{60}{210} \approx 2.85\text{ V}$$

Since $2.85\text{ V} > 0.7\text{ V}$, the diode must be **ON (Conducting)**.

2. Calculate currents:

Because the diode is ON and in parallel with R_2 , it clamps the voltage across R_2 to its threshold voltage: $U_{R2} = V_D = 0.7\text{ V}$. The total current leaving the source (I_{total}) passes through R_1 :

$$I_{total} = \frac{E - V_D}{R_1} = \frac{10 - 0.7}{150} = \frac{9.3}{150} = 0.062\text{ A} = 62\text{ mA}$$

The current flowing through the resistor R_2 is:

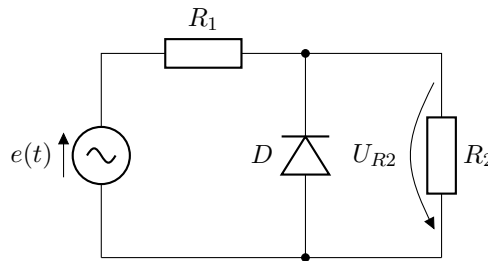
$$I_{R2} = \frac{V_D}{R_2} = \frac{0.7}{60} \approx 0.01167 \text{ A} = 11.67 \text{ mA}$$

By Kirchhoff's Current Law, the current through the diode I_D is:

$$I_D = I_{total} - I_{R2} = 62 - 11.67 = \mathbf{50.33 \text{ mA}}$$

Exercise 6: AC Source with Diode

The DC source is replaced by $e(t) = 5 \sin(\frac{2\pi}{T}t)$ with $T = 20 \text{ ms}$. Let $R_1 = R_2 = R$. The diode is placed in parallel with R_2 , pointing upwards (cathode to signal, anode to ground) as per the lecture notes.



Analysis over one period T :

- **Positive Half-Cycle** ($t \in [0, 10 \text{ ms}]$): $e(t) > 0$. The potential at the top node is positive relative to ground. The diode is reverse-biased (**Blocked**). The circuit acts as a simple voltage divider.

$$U_{R2}(t) = \frac{R_2}{R_1 + R_2} e(t) = \frac{1}{2} \times 5 \sin(\omega t) = \mathbf{2.5 \sin(\omega t)}$$

- **Negative Half-Cycle** ($t \in [10 \text{ ms}, 20 \text{ ms}]$): $e(t) < 0$. The potential at the top node becomes negative, making the anode (ground) more positive than the cathode. The diode is forward-biased (**Passing/ON**). It clamps the voltage across R_2 to its negative threshold:

$$U_{R2}(t) = -V_D = \mathbf{-0.6 \text{ V}} \quad (\text{As per the whiteboard notes})$$

This creates a clipped half-wave sinusoidal output.

Exercise 7: Zener Diode Datasheet

Given Zener specifications: $V_Z = 12 \text{ V}$ at $I_Z = 50 \text{ mA}$, $I_{Zmin} = 0.5 \text{ mA}$, $I_{Zmax} = 100 \text{ mA}$, and $R_Z = 20 \Omega$.

1. Calculate the voltages V_{Zmin} and V_{Zmax}

The real Zener diode is modeled by a threshold voltage V_{Z0} in series with its dynamic resistance R_Z . First, find V_{Z0} using the nominal values:

$$V_Z = V_{Z0} + R_Z I_Z \implies 12 = V_{Z0} + (20 \times 0.050) \implies 12 = V_{Z0} + 1 \implies V_{Z0} = 11 \text{ V}$$

Now calculate the extremes:

$$V_{Zmin} = V_{Z0} + R_Z I_{Zmin} = 11 + (20 \times 0.0005) = 11 + 0.01 = \mathbf{11.01\text{ V}}$$

$$V_{Zmax} = V_{Z0} + R_Z I_{Zmax} = 11 + (20 \times 0.100) = 11 + 2.0 = \mathbf{13.00\text{ V}}$$

2. Equation for the characteristic line

The voltage-current equation in the avalanche/breakdown region ($I_{Zmin} \leq I_Z \leq I_{Zmax}$) is:

$$\mathbf{V_Z = 11 + 20I_Z}$$

3. Calculate V_{Z1} and V_{Z2}

For $I_{Z1} = 20\text{ mA}$:

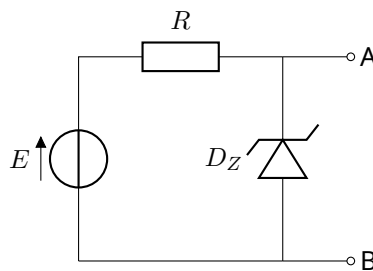
$$V_{Z1} = 11 + (20 \times 0.020) = 11 + 0.4 = \mathbf{11.4\text{ V}}$$

For $I_{Z2} = 80\text{ mA}$:

$$V_{Z2} = 11 + (20 \times 0.080) = 11 + 1.6 = \mathbf{12.6\text{ V}}$$

Exercise 8: Zener Applications

Given: $R = 1\text{ k}\Omega$, ideal Zener breakdown $V_Z = 6.2\text{ V}$, forward bias drop $V_D = 0.6\text{ V}$.



1. Determine U_{AB} for varying positive E (Zener Regulator mode)

The Zener is placed in reverse bias (cathode facing the positive potential).

- **(a) For $E = 12\text{ V}$:** Since $12\text{ V} > 6.2\text{ V}$, the Zener diode undergoes breakdown. It regulates the voltage across nodes A and B.

Result: $U_{AB} = V_Z = 6.2\text{ V}$

- **(b) For $E = 5\text{ V}$:** Since $5\text{ V} < 6.2\text{ V}$, the Zener diode does not reach breakdown. It acts as an open circuit (OFF). No current flows through R , so there is no voltage drop across R .

Result: $U_{AB} = E = 5.0\text{ V}$

2. Determine U_{AB} for varying polarities of E

- **(a) For $E = 5\text{ V}$:** As established above, the Zener is in reverse bias but below the breakdown threshold. It is blocked.

Result: $U_{AB} = 5.0\text{ V}$

- **(b) For $E = -12\text{ V}$:** The voltage source polarity is reversed. This forward-biases the Zener diode (current pushes from anode to cathode). In forward bias, a Zener diode behaves exactly like a standard silicon diode with a forward voltage drop V_D .

Result: $U_{AB} = -V_D = -0.6\text{ V}$